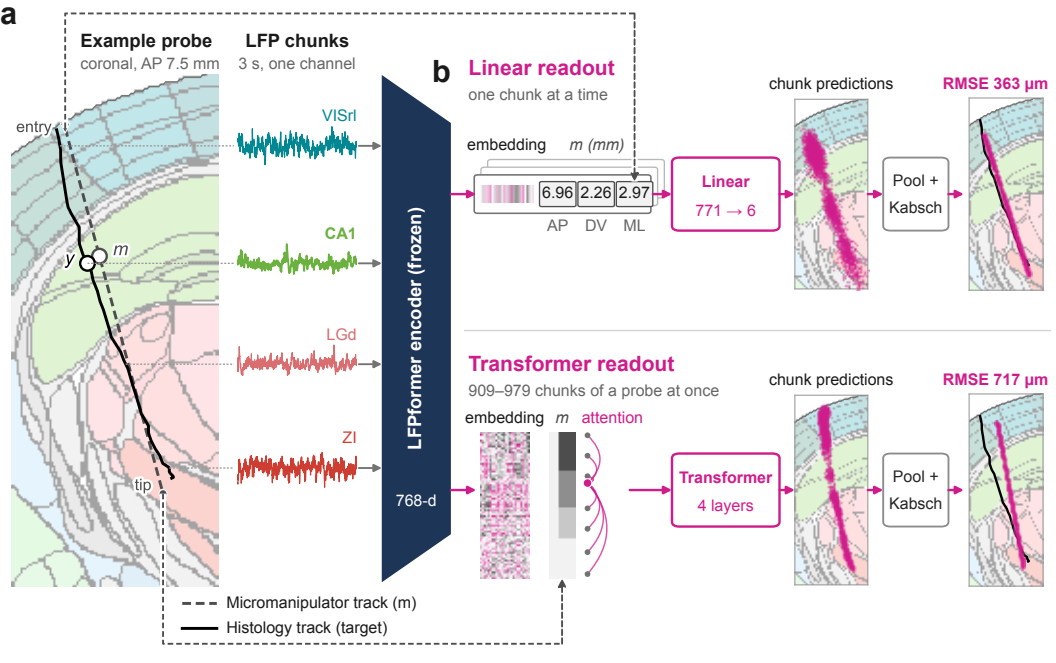

a
Example probe
coronal, AP 7.5 mm
LFP chunks
3 s, one channel
entry
y
m
tip
VISrl
CA1
LGd
ZI
LFPformer encoder (frozen)
768-d
b
Linear readout
one chunk at a time
embedding
m (mm)
6.96
2.26
2.97
AP
DV
ML
Linear
771 → 6
chunk predictions
Pool +
Kabsch
RMSE 363 µm
Transformer readout
909–979 chunks of a probe at once
embedding
m
attention
Transformer
4 layers
chunk predictions
Pool +
Kabsch
RMSE 717 µm
Micromanipulator track (m)
Histology track (target)